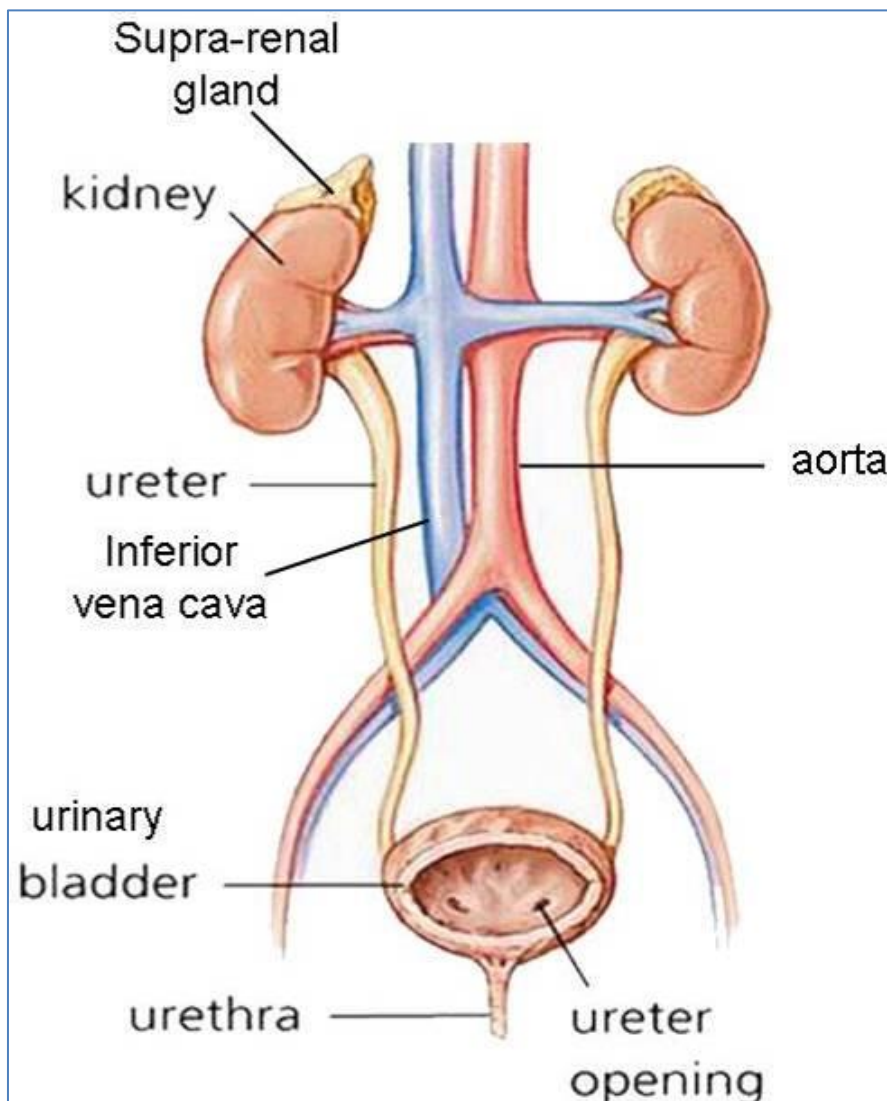


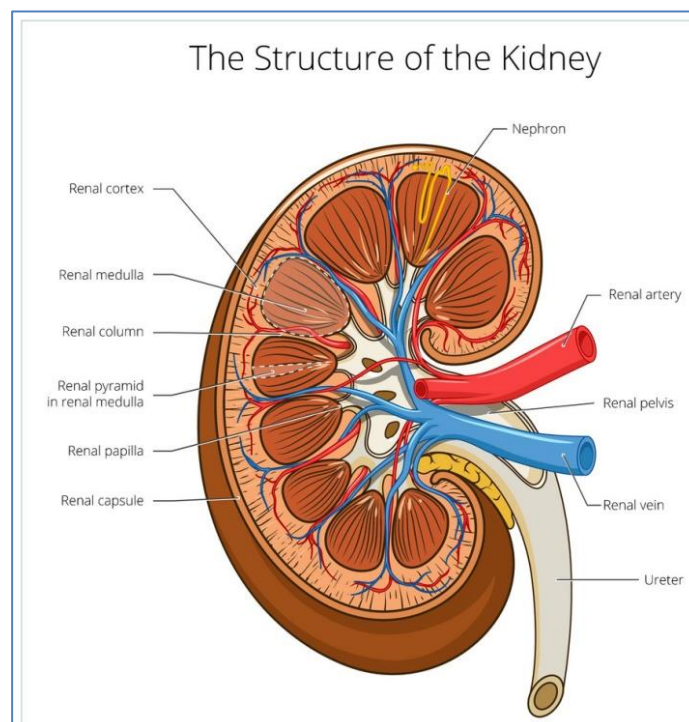
URINARY SYSTEM

- It is the system responsible for excretion of toxic metabolic waste products (like urea, creatinine and nitrogen), excess salt and water from the blood through formation of urine to be excreted to the outside through urinary duct system.
- The anatomy of urinary system consists of four parts;
 1. Two kidneys,
 2. Two ureters,
 3. Urinary bladder
 4. Urethra.



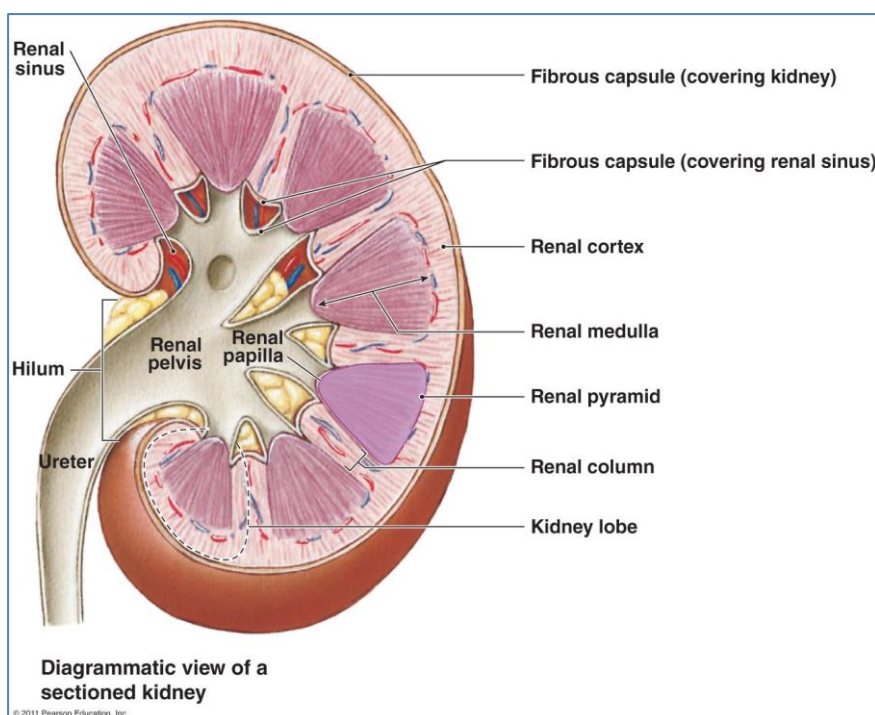
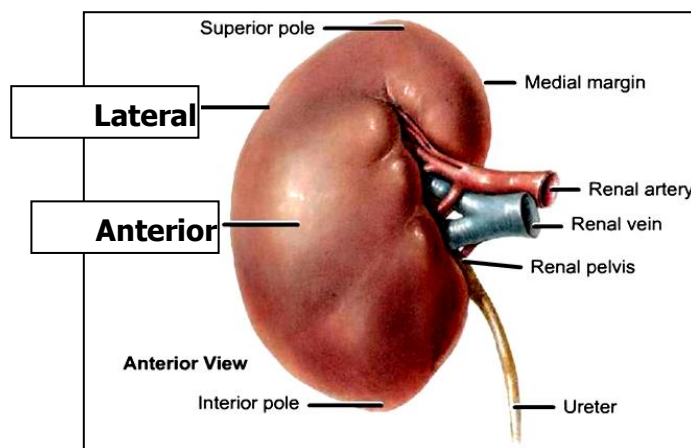
Kidney

- Right and left kidneys are present on the **upper part of the posterior abdominal wall** extends between **T12 - L3** vertebrae.
- **Shape:** Kidney is **bean** shaped, it has:
 - 2 ends: upper and lower ends.
 - 2 borders: lateral convex border and medial concave border containing the hilum.
 - 2 surfaces: anterior and posterior surfaces.
 - **Hilum of kidney:** the site where the renal vein, renal artery and renal pelvis enter or leave the kidney. They are present in the hilum and arranged as renal vein anteriorly, renal artery in the middle and renal pelvis posteriorly.
- **Size:** 12 cm length x 6 cm width x 3 cm thickness.
- **Peritoneal relations:** it is retroperitoneal organ. The peritoneum covers the anterior surface of the kidney.
- It moves with respiration as it is related to the diaphragm.



- **Internal structure of the kidney**

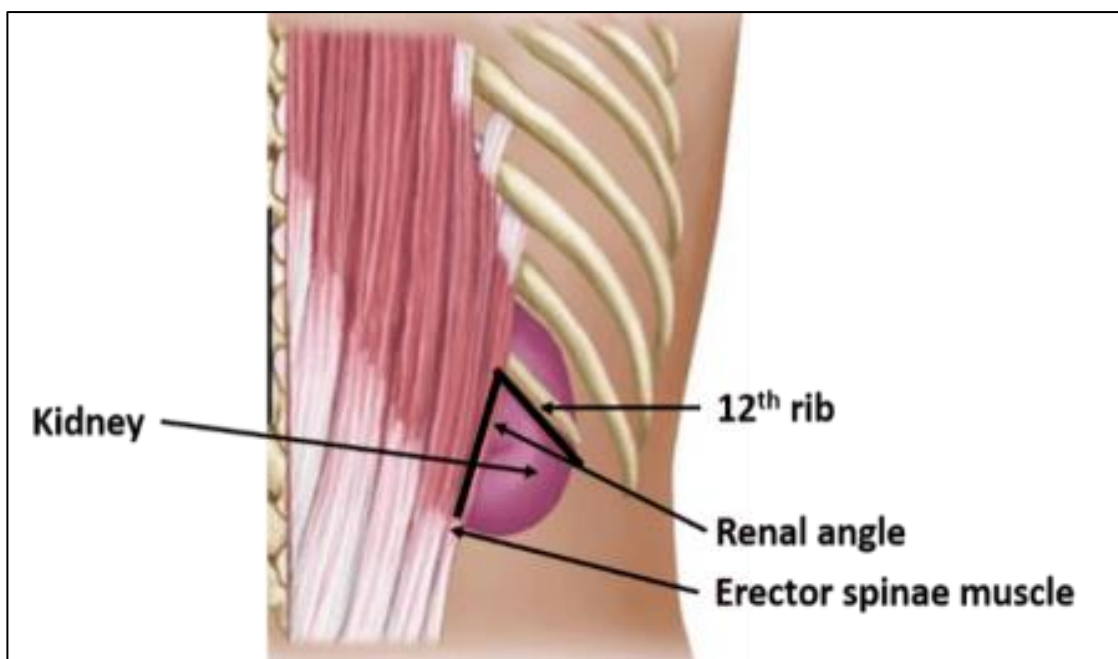
- The renal pelvis **is funnel in shape at the upper end of ureter.**
- **The renal pelvis receives 2-3 major calyces, each of which receives 2-3 minor calyces.**
- **Each minor calyx is indented by a renal papilla, the apex of renal pyramid, from which the urine is excreted.**
- **The renal pyramids with the overlying renal cortex form the lobes of the kidney.**



Applied anatomy

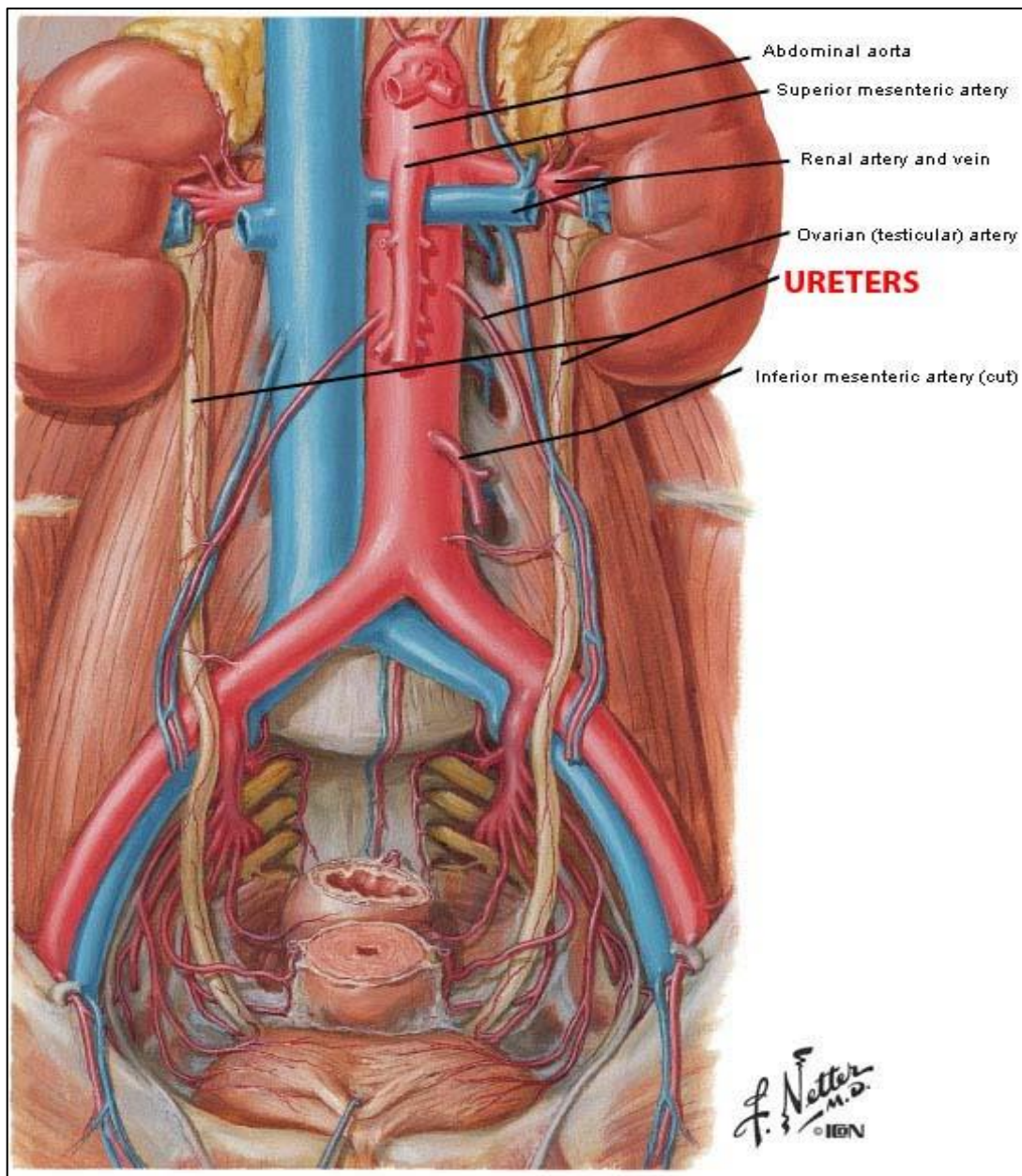
- Renal angle

1. It is the angle between the lower margin of the 12th rib and lateral margin of the erector spinae muscle.
2. At this angle the kidney lies close to the body surface.
3. Renal pain is felt as dull aching pain in renal angle.
4. Perinephric abscess causes swelling and tenderness at renal angle.



Ureter

- It is a narrow muscular tube about **25 cm** long, **6 mm** in diameter.
- It **extends** from the pelvi-ureteric junction to the postero-superior angle of urinary bladder and conducting urine from the kidney to urinary bladder.
- It is formed of **abdominal** part, **pelvic** part and **intramural** part.
- It has **3 sites of constrictions**: at pelvi-ureteric junction (beginning), at its termination (intra-mural part) and one constriction in between (site of crossing over the inlet of pelvis).

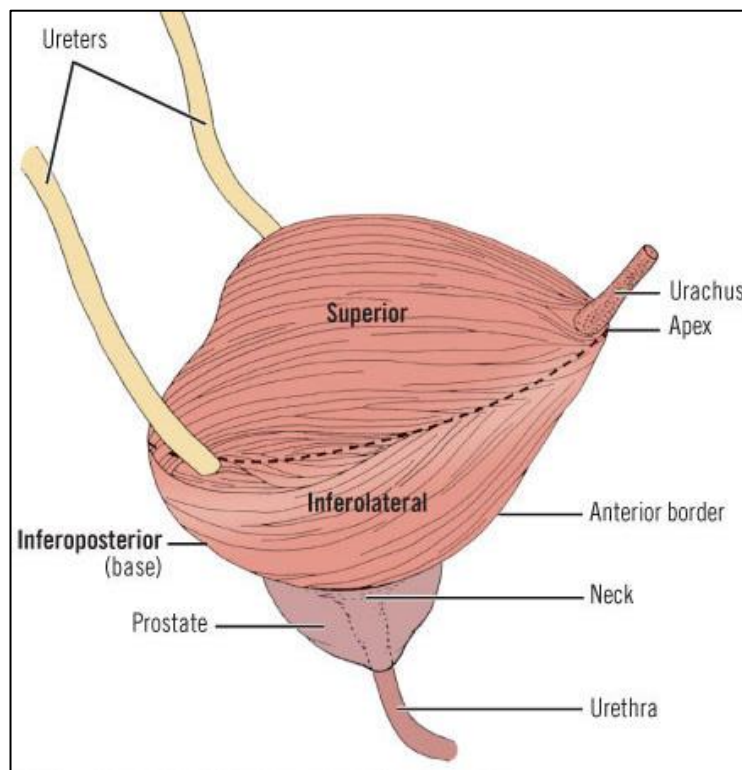


- **Applied anatomy**

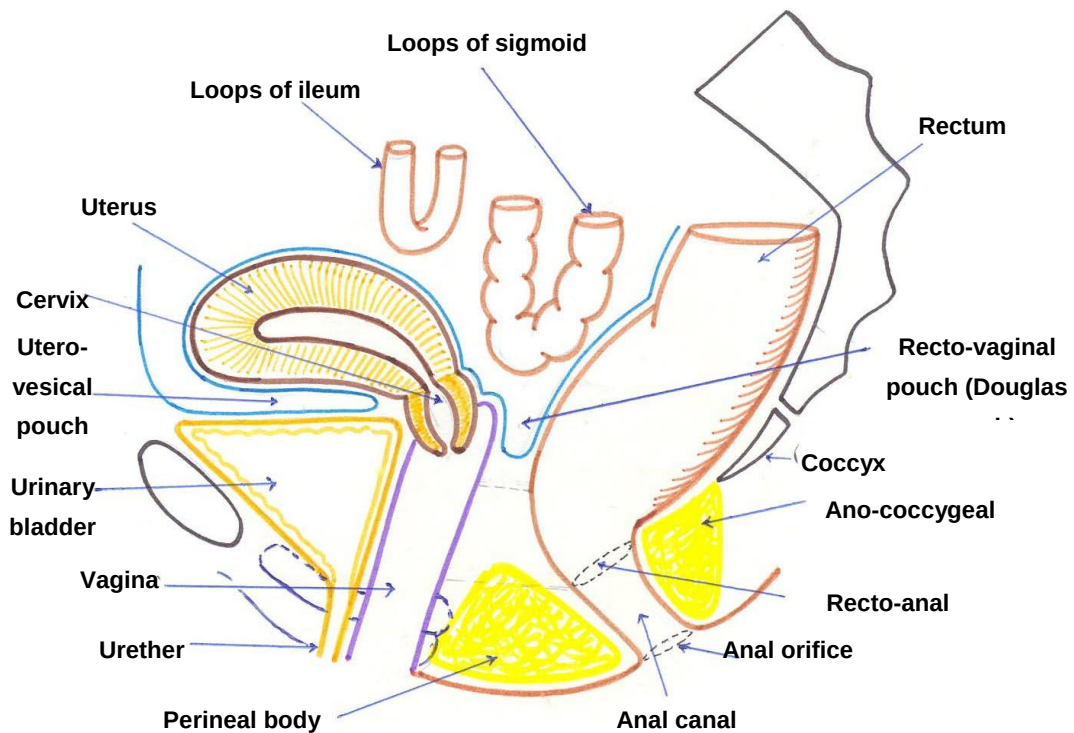
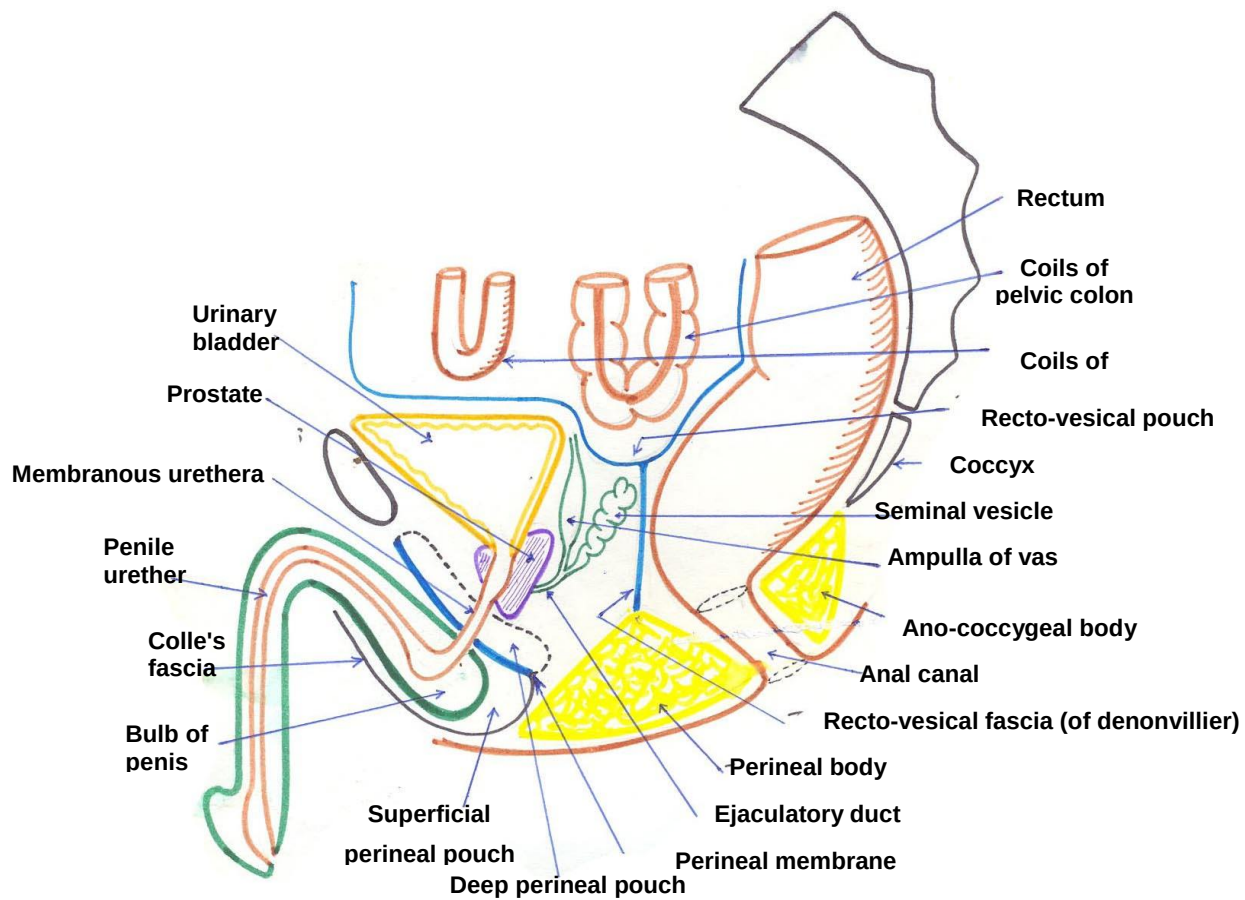
Renal stones in the ureter may be impacted at the sites of normal ureteric constrictions. This leads to severe colicky pain.

Urinary bladder

- It is a hollow muscular organ acts as reservoir of urine with a maximum capacity of 500 ml. The empty bladder is pyramidal in shape having three sides.
- **Site:** it is present inside the anterior part of pelvic cavity.
- It **receives** the 2 ureters and the urine leaves it through the urethra.
- It has **4 surfaces:** superior, 2 infero-lateral and posterior (base of bladder) surfaces.
- It has **4 angles:** anterior, 2 postero-superior and inferior (bladder neck) angles.
- The bladder neck is surrounded by the **base of the prostate**.
- The base of bladder is related to ampulla of vas deferens and seminal vesicles in males and vagina in females.



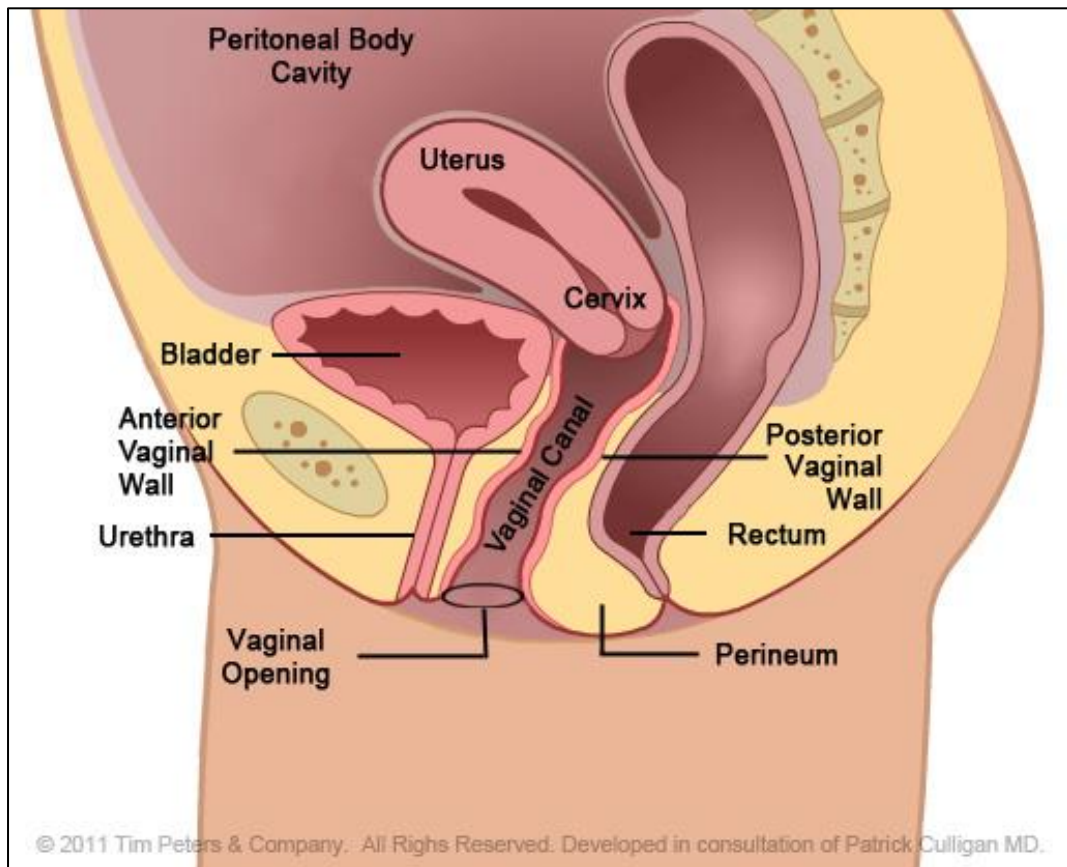
Sagittal section in Male Pelvis



Sagittal section in Female Pelvis

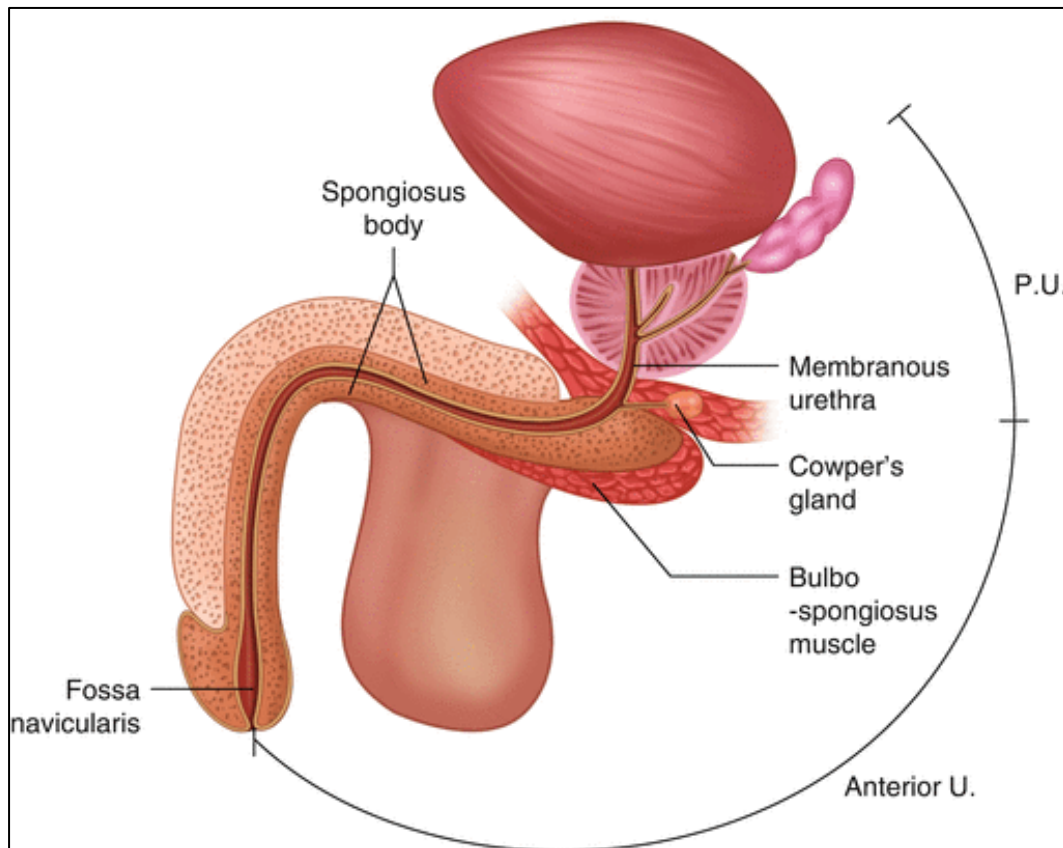
Urethra

- It is a muscular tube which carries the urine from the urinary bladder to outside the body.
- **Beginning & termination:** It ***starts*** at the neck of the urinary bladder, as the **internal urethral opening**, and ***ends*** externally as the **external urethral opening**.
- ★ **Female urethra:** short and wide tube about 4 cm long, embedded in the anterior wall of vagina. It is directed downwards and forwards

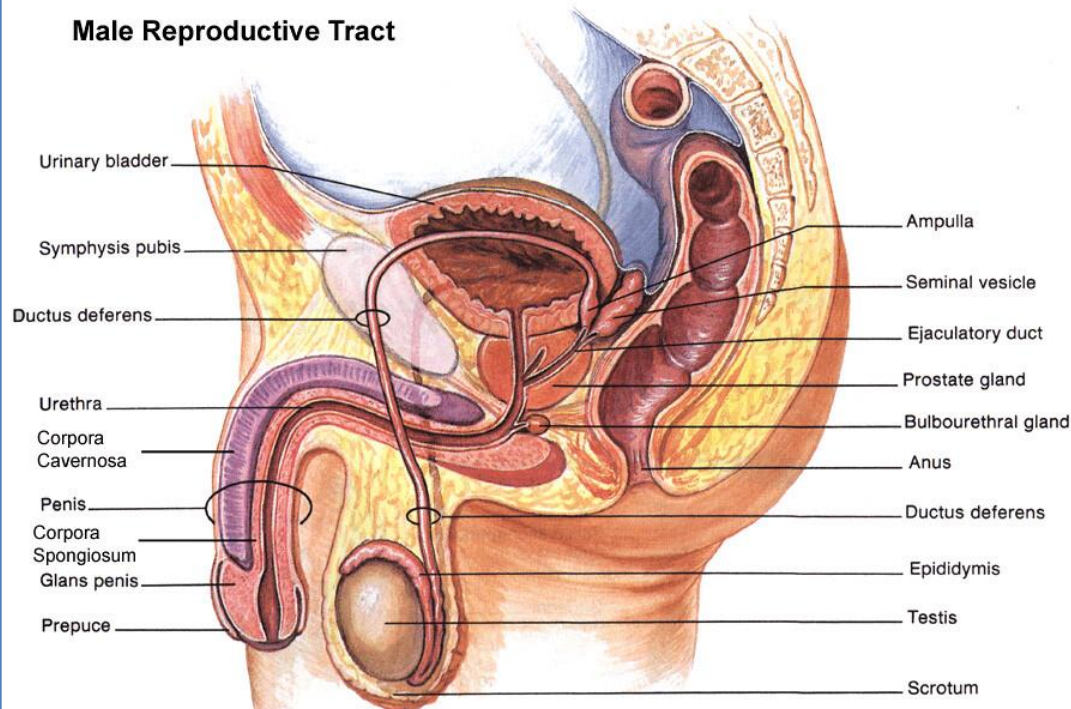


★ Male urethra

- The male urethra is **20 cm** long and **extends** from the internal urethral meatus (orifice) in the urinary bladder (surrounded by involuntary internal urethral sphincter), to the external urethral meatus at the tip of the penis.
- It is **S shape**, in the flaccid penis.
- **Function**
 - It is a **common duct** for both urinary and genital systems.
 - It allows passage of **urine and semen** to the outside.
- It is formed of 3 parts:
 - a- **Prostatic urethra**
 - It is the **widest** part, about 3 cm long, passes within the prostate.
 - Its posterior wall receives the 2 ejaculatory ducts and openings of the prostatic glands.
 - b- **Membranous urethra**
 - The **narrowest** part about 2 cm long.
 - **It is** surrounded by **the voluntary external urethral sphincter muscle**
 - c- **Penile urethra**
 - The **longest** part about 15 cm long lies within the penis.
 - It **ends** at the **external urethral meatus**.



Male Reproductive Tract



Modified from Van De Graaff, *Human Anatomy*, Wm. C. Brown: Dubuque, IA, 1988.